

## **COLDSTART: Anticipating health-system crises before the outcome is observable**

### **1. Summary**

When a health crisis begins, the quantity decision-makers need to forecast (presentations, demand, occupancy) is precisely the one for which almost no local outcome data yet exist: the outcome series is a handful of noisy observations, so systems that perform well on years of local history are weakest when early decisions matter most.

There is another source of quantitative information: published evidence from analogous events elsewhere. It is rarely tested as formal prior information, because of the two steps that matter most: **reliable extraction, and transport to the new setting**. A confidently wrong prior is worse than no prior: it is most influential exactly when local data cannot correct it.

This project asks the central question directly: **can published quantitative evidence provide useful information when local outcome data are insufficient at crisis onset, and can we detect early when it should not be trusted?** **LiteRev-Evidence**, the prototype I built, already walks this pipeline from a plain-language question to certainty-graded extraction, pooled parameter distributions and tuned candidate models. The project turns the prototype into a validated instrument and tests it: I will (1) establish the error structure of automated extraction and propagate it into evidence-derived priors, for parameter values and for the rest of what the literature carries (predictor variables and lags, model forms, outcomes and their thresholds), each transported to the local setting; (2) develop an uncertainty-aware latent-regime framework for health-system escalation, with those priors and a short-window resilience signal; (3) test, by strict rolling-origin reconstruction of each historical moment's information, whether the evidence improves cold-start forecast skill, how fast that value decays, and when it causes harm; and (4) determine whether any gain is large enough to change operational decisions. The decisive experiment runs a pre-registered library of candidate models (local baselines, surveillance algorithms, machine-learning and latent-regime models; mechanistic transmission components only where the crisis is epidemiological) through an automated engine that checks each model's assumptions, scores it on established metrics, and selects each arm's champion under rules fixed in advance. Exactly one contrast is confirmatory: the evidence-informed champion against the local-only champion, repeated once, sequentially, as the heat generalisation test.

Every component is validated first on open surveillance and demand series (a public, re-runnable benchmark), then on Geneva's operational series. Respiratory epidemics carry the confirmatory test, heatwaves the sequential generalisation test, Geneva legionellosis a scoped year-4 extension. The outcome is intentionally falsifiable: if literature-derived evidence improves cold-start forecasts, the project establishes when and by how much; if not, or if harm cannot be detected early enough, a boundary condition for automated evidence-informed forecasting. Either way: a validated answer and an open framework. For event types whose models validate, the framework runs by month 48 as a **daily observation-mode dashboard** with Geneva's emergency services, forecasts and validity checks refreshed automatically, recorded, never yet steering care: the nucleus of a living surveillance platform.

The project rests on an unusual combination: fifteen years of quantitative-finance work on rare-event risk and regime models, LiteRev-Evidence, and Swiss emergency data. Ambizione turns these into an independent research programme, led by me, at the intersection of evidence synthesis, quantitative modelling and emergency public health.

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## 2. Proposed research

### 2.1 Current state of research in the field

#### 2.1.1 Forecasting health-system crises: strong in steady state, weak at onset

Syndromic surveillance detects departures from expected baselines [Farrington 1996; Noufaily 2013], and forecast ensembles perform strongly in data-rich settings [Cramer 2022; Sherratt 2023]. Dispatch records lead confirmed surveillance: the 2009 A(H1N1) autumn onset was identified eight days in advance in three European regions [Rosenkötter 2013], long call series track influenza-like illness [Bonora 2025], and our systematic review maps the field [Edjinedja 2026].

The decisive limitation is **history dependence**: data-adaptive models need enough local observations to learn seasonality, weather response and crisis dynamics; reviews of emergency-demand forecasting catalogue which model families perform for which outcome and horizon [Wargon 2009], but every recommended model is trained on years of local history.

**The cold-start problem is specifically a labelled-outcome problem.** At crisis onset, context variables abound but the outcome to be forecast does not, and abundance itself can mislead: Google Flu Trends overestimated influenza activity by more than a factor of two [Lazer 2014]. Early signals do not close the gap: wastewater needs a shedding-to-incidence relationship that itself comes from external evidence; transfer learning needs comparable contemporaneous observations. The common issue is **how to use external quantitative information without pretending it is perfectly transferable.**

#### 2.1.2 The unused resource: published evidence as quantitative prior information

Bayesian borrowing is well developed: power priors [Ibrahim 2000], commensurate priors [Hobbs 2011], meta-analytic-predictive priors with robust conflict protection [Schmidli 2014]. Informative priors have been used to forecast disease under sparse local data [Cook 2023]. What has not been established is narrower and harder: whether an **automatically constructed** evidence prior, with extraction error and transportability uncertainty propagated into it, improves **operational cold-start forecasts under strict historical information constraints**, and whether harmful borrowing is detected early enough to act.

**Parameter values are not the only transferable object.** The literature also carries **predictor variables and lag structures, model-form evidence** (which algorithm families work for which crisis type and horizon) and **outcome and threshold definitions**. Living systematic reviews keep syntheses current [Elliott 2014], yet no operational forecasting framework consumes these objects systematically.

**Extraction.** Automated extraction makes large-scale synthesis feasible, but numerical extraction remains less reliable than categorical (reported accuracy roughly 47–88% versus 74–96%), and omissions dominate errors [Shankar 2026]. The question is whether extraction and pooling preserve the dispersion a calibrated prior needs.

**Transportability.** Even perfectly extracted estimates may not transfer across populations, case definitions, health systems or policy regimes. Formal transportability tools exist [Bareinboim 2016; Dahabreh 2020; Degtiar 2023] but are not integrated into any operational cold-start framework.

### 2.1.3 Representing escalation as a state, not only a point forecast

Emergency operations are interested in state (routine, elevated, strained, critical) rather than only tomorrow's count. Latent-state representations are established in surveillance: Poisson hidden Markov models distinguish epidemic from non-epidemic periods [Le Strat 1999; Watkins 2009]; extreme-value methods have been applied to Swiss hospital visits [Coles 2001; Ranjbar 2022]. Here they serve one purpose: a latent ordinal state as the **common representation for testing evidence borrowing at crisis onset**, with an extreme-value component for rare critical exceedances. Mechanistic transmission models [Keeling 2008] enter as components only where the crisis is epidemiological; environmental crises such as heat are driven by exposure-response, not transmission. Critical-slowness theory (rising variance and lag-1 autocorrelation before some transitions [Scheffer 2009]; epidemic applications [O'Regan 2013; Brett 2018; Southall 2021]) uses the shape of a short recent series: a **secondary information channel**, tested empirically.

### 2.1.4 From forecast accuracy to decision value

Proper scoring rules reward calibration [Gneiting 2007], but a calibrated forecast is irrelevant if it changes no decision; decision-analytic methods evaluate predictions under explicit consequences [Vickers 2006]. 72-99% of reported clinical alarms have been false or non-actionable [Winters 2018]; mortality-calibrated heat thresholds need not match demand thresholds [Lung 2021]; pollution-asthma emergency associations lack recent Swiss acute-episode studies [Zheng 2015]. Methods are therefore evaluated first statistically, then under an elicited operational loss structure.

### 2.1.5 The specific gap addressed by this project

Ongoing programmes occupy the neighbouring ground: multi-model forecasting hubs run routinely (ECDC RespiCast, US CDC FluSight), Horizon Europe funds epidemic-intelligence infrastructure (including GeoAI4EI, in which I take part), and the Franco-Swiss GESICA programme builds cross-border crisis intelligence. All develop surveillance, forecasting and decision-support capability from local data; none has as a stated objective testing whether automatically synthesised external evidence should enter a forecast as formal prior information at local cold start.

What has not been established is whether these pieces connect around the **cold-start question**, and whether the loop from living evidence synthesis to variable choice, model choice, thresholds and automatically re-validated forecasts can be closed under pre-registered rules.

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## 2.2 Current state of personal research and competences required for the project

My route is unusual and is why the project is tractable: fifteen years in quantitative finance (risk modelling, extreme-value estimation, regime models, stress testing), then a doctorate in biomedical sciences at Geneva (defended 18 December 2023).

### 2.2.1 Prediction under sparse and imperfect information

In [Orel 2022] I predicted individual HIV status from socio-behavioural characteristics across East and Southern Africa, establishing where models transported between countries. [Merzouki 2021] and [Esra 2023] developed the same theme. I now **senior-author** that line (last author, [Ng'ambi 2026], accepted: machine-learning classification of cardiovascular disease history across harmonised WHO

STEPS surveys): this proposal's transportability problem, met first elsewhere, moved from conducting to directing.

### 2.2.2 Automated evidence synthesis

In [Orel 2023] I introduced **LiteRev**, an automated literature-review tool: natural-language or Boolean query across the open-access databases, deduplication, corpus mapping, iterative relevance suggestion. I led its development with **Aziza Merzouki** (PhD, computer science) and secured development funding on my own initiative: **CHF 30,000** (UNIGE), **CHF 10,000** (Venture Kick), **CHF 20,000** (Mimosa), outside any group grant. The AI-in-EMS systematic review [Edjinedja 2026], within GESICA, used it in its published methods.

### 2.2.3 Outbreak and health-system modelling in Switzerland

In [Orel 2024] I compared clinical severity between Delta and Omicron sub-lineages in a Swiss tertiary centre. [Estill 2020] built age-structured scenario models for the Swiss SARS-CoV-2 epidemic under planning time pressure: the experience this proposal's question comes from. I contributed to WHO African Region reporting and seroprevalence estimation [Nwosu 2021].

### 2.2.4 The prototype and test bed: LiteRev-Evidence

Since 2024 I have developed **LiteRev-Evidence**, extending LiteRev into structured quantitative extraction and modelling: the full review pipeline in working form, from a query searched live across the open literature, through deduplication, PRISMA-accounted screening and PICO extraction, to **provenance-tracked extraction graded for study-design certainty and pooled into parameter distributions**, and onward to the model step: candidate predictor variables with their supporting articles; typed outcome definitions with ready-made emergency-demand targets (ED overload, bed occupancy, call surge); **model fitting across algorithm families with automated hyperparameter search, leakage-safe time-series validation and ranking on a stated metric**; and compartmental components, parameterised from extracted ranges, that can re-enter the statistical model as predictors. Per-scenario **living reviews re-run on schedule**, every update previewed before it is applied: the evidence-to-model loop this proposal interrogates, in prototype form.

For GESICA I built the Geneva-Vaud-Neuchâtel data foundation: **77 notifiable diseases in eight model classes**, and a referenced inventory of **28 surveillance sources (23 infectious, 5 environmental/non-infectious)** each documented for institution, coverage, resolution, latency, access and quality. It is why validation domains are chosen by **model class**, and why the proposal rests on a mapped, not assumed, data landscape.

**The prototype makes the research feasible rather than aspirational**: the fit-tune-rank mechanics T3.3's selection engine registers already run in embryonic form, so the grant adds the science (the two arms, the registered rules, episode-level inference), not the plumbing; the validated instrument it becomes is itself a deliverable.

### 2.2.5 Linked data on a contrasting crisis archetype

I lead the data work on a cantonal study **already under way** (BASEC 2026-00324, ethics granted) linking confirmed legionellosis cases in Geneva to individual domestic hot-water installations: a linkage to my knowledge unique, supplying a contrasting waterborne archetype.

**Competences.** Bayesian hierarchical and regime-switching estimation, extreme-value modelling, stress testing under misspecification, forecast evaluation and decision analysis come from the risk

work and form the core of WP2-WP3; extraction and pooling from LiteRev-Evidence; domain knowledge from the doctorate, the Swiss COVID-19 work and GESICA.

## 2.3 Detailed research plan

### 2.3.1 Objectives and hypotheses

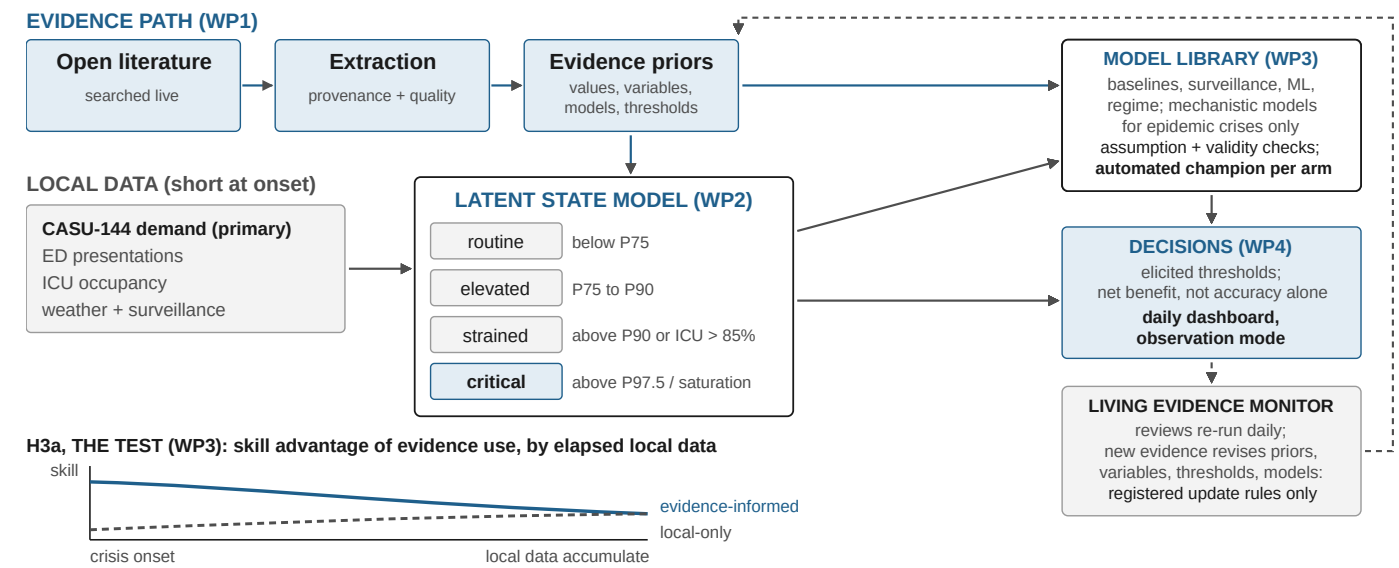


Figure 1: cold-start problem, evidence pipeline, model library with automated selection, living update loop.

To determine **whether, and under what conditions, published quantitative evidence provides useful information when local outcome data are insufficient at the onset of a health-system crisis, and whether the resulting forecasts change decisions.**

**One central hypothesis, H3a;** everything else is subordinate to it:

|     | Role               | Statement                                                                        |
|-----|--------------------|----------------------------------------------------------------------------------|
| H1  | Validation         | Can the evidence be trusted enough to use?                                       |
| C2  | Adequacy criterion | Is the state representation fit to compare borrowing in?                         |
| H3a | Central hypothesis | <b>Does systematically extracted evidence improve cold-start forecast skill?</b> |
| H3b | Robustness         | Is adaptive borrowing safe when the prior is wrong?                              |
| H3c | Secondary channel  | Do resilience indicators add information beyond the evidence?                    |
| H4  | Decision value     | Is the gain large enough to change an operational choice?                        |

*O1: Make published evidence usable without hiding its uncertainty*

**H1.** Automated extraction will systematically **understate the dispersion of the evidence-derived distribution** (reported within-study uncertainty and between-study heterogeneity), producing priors that are too concentrated; an explicit measurement-error layer will recover enough of the missing dispersion to construct usable priors.

Omissions dominate extraction errors [Shankar 2026] and shrink dispersion systematically; if uncorrectable, the project establishes a boundary condition. The evidence target spans four object classes (parameter values; predictor variables and lags; model forms; outcome and threshold definitions: T1.1); H1 is tested on the quantitative core, and classes 2 to 4 are validated externally against published syntheses (T1.2).

*O2: Represent escalation in a form that separates state from the point forecast*

**C2, model adequacy criterion.** The latent-state representation must yield **identifiable** parameters and **calibrated** escalation-state probabilities at matched false-alarm rates. Its role is the common state representation in which borrowing strategies are compared, not a claim that regime switching generally beats thresholding a point forecast.

T2.1's identifiability study and T3.3's calibration checks either establish adequacy or trigger the ordinal fallback; either outcome leaves H3a intact. Extreme-value modelling covers the critical tail; critical-slowness indicators are **supporting covariates**.

*O3: Test the cold-start hypothesis and map failure*

**H3a.** An evidence-informed forecasting configuration (evidence-derived priors, variable sets and model forms) improves probabilistic forecast skill during the early phase of a crisis, with the advantage declining as local observations accumulate.

**H3b.** Adaptive borrowing that discounts the evidence when prior-data conflict emerges is **non-inferior** to fixed borrowing under well-specified priors, within a pre-specified margin  $\Delta$  on the CRPS skill score, and is **superior** to fixed borrowing under deliberately misspecified priors.

$\Delta$  is fixed at the second registration point, justified against the evidence-informed versus local-only effect the study is powered to detect.

**H3c.** Resilience indicators add predictive information beyond the evidence-derived prior and the local level/trend signal when the outcome history is short.

**Primary confirmatory comparison (one, stated once):** the **evidence-informed champion** against the **local-only champion**, both produced by the identical pre-registered automated procedure (library, validity checks, selection rule: T3.3), by **CRPS skill score**, over the pre-specified cold-start window, **on respiratory episodes only**. Heat repeats the identical contrast as a **sequential generalisation test**, run only if the respiratory test is met; the fixed order controls the family-wise error rate. The **shape of the advantage over elapsed local data** is reported: it should decay to nothing.

*O4: Establish whether predictive improvement is decision-relevant*

**H4.** Decision-analytic evaluation under the losses and escalation thresholds of emergency responders can rank modelling strategies differently from generic accuracy criteria; the evidence-derived strategy is useful only when its gain crosses a decision threshold.

These are downstream tests of value; observation mode is an extension, not a prerequisite.

*Validation domains*

The domains span **two contrasting model classes** (GESICA classification). Both arms score **the same quantity, built the same way**: daily emergency demand from the CASU-144 series, restricted to the cause classes the archetype's evidence base concerns (T3.0): respiratory-related demand for epidemics; **heat-sensitive demand** for heat, since Swiss evidence places heat effects in dehydration, renal and psychiatric admissions, with a weak respiratory effect at older ages [Schulte 2024; Ragettli 2019].

| Archetype                   | Role                 | Dynamics                            | Outcome and data                                     |
|-----------------------------|----------------------|-------------------------------------|------------------------------------------------------|
| <b>Respiratory epidemic</b> | Primary confirmatory | Transmissible, multi-wave, seasonal | Respiratory-related demand; COVID-19, influenza, RSV |
| <b>Heatwave</b>             | Sequential           | Environmental, short,               | Heat-sensitive demand, same                          |

| Archetype                  | Role             | Dynamics                     | Outcome and data                             |
|----------------------------|------------------|------------------------------|----------------------------------------------|
| <b>Waterborne outbreak</b> | generalisation   | sharply peaked               | construction; MeteoSwiss exposures           |
|                            | Year-4 extension | Common-source, environmental | Geneva legionellosis linked to installations |

### *What the project does not claim*

It does **not** aim to outperform forecast hubs in the data-rich regime, assume literature-derived priors are beneficial, claim critical slowing down as universal, or promise a clinically steering alarm system by month 48: the dashboard runs strictly in observation mode. The contribution is narrower: **whether accumulated quantitative evidence can earn a formal role in forecasting before local outcome data become informative, and a map of when it should not be trusted.**

## 2.3.2 Work packages and methods

### *WP1: From published evidence to usable priors (M1-M20)*

#### *T1.1: Define the evidence target and register the protocol (M1-M4)*

**The evidence target spans four object classes;** inclusion criteria, effect measures, uncertainty representation and transportability variables are fixed before extraction begins. **(1) Parameter values, the quantitative core (benchmarked in full):** lag-structured **weather-demand coefficients** (daily mean temperature, heat-day exceedance, ozone, PM10); **surge magnitude and timing** (peak-to-baseline ratio, time to peak, onset growth rate); **length-of-stay and occupancy distributions** (ward and ICU); **admission fractions;** transmission parameters ( $R_0/R_t$ , serial interval) and shedding-to-incidence conversions secondary, off the critical path. **(2) Predictor variables and lag structures** reported as informative per crisis type. **(3) Model-form evidence:** which algorithm families the forecasting literature finds effective for which outcome, horizon and crisis type [Wargon 2009], feeding T3.3's library. **(4) Outcome and threshold definitions:** which series are used operationally and at what levels warnings fire [Lung 2021], entering T2.1's anchors and T4.1's elicitation as evidence-based starting points adapted to the local setting. Classes 2 to 4 carry provenance and certainty grades; H1's benchmark applies to class 1.

#### *T1.2: Build the quantitative extraction benchmark (M3-M9)*

Two independent expert extractors extract target quantities from a stratified random sample of **300** publications, with adjudication; stratification covers parameter class, reporting quality, study design and **source type** (journal article versus timestamped situational report), so error is characterised separately for T3.2's faster secondary sources. An **open external layer** scores retrieval against the reported inclusion sets of published systematic reviews in the target classes, and extraction and pooling against their published summary estimates. The benchmark is an open deliverable.

#### *T1.3: Characterise automated extraction error (M7-M14)*

Compare automated extraction with the benchmark on point estimates, uncertainty, omissions and between-study dispersion, with sensitivity to the underlying model version.

#### *T1.4: Construct uncertainty-aware evidence distributions (M12-M18)*

Represent extraction error explicitly as measurement error; compare inverse-variance and quality-weighted pooling, meta-analytic-predictive priors and power-prior discounting. Output: a prior with provenance, extraction uncertainty and a documented borrowing weight.

### T1.5: Transportability screen (M15–M20)

Characterise effect modifiers and study-setting differences relevant to Geneva; where transport is weak, widen or discount the prior, producing the metadata WP3's conflict analysis needs.

**Deliverables.** D1.1 open benchmark; D1.2 error analysis; D1.3 evidence-to-prior library.

**Risk.** If expert time binds, the benchmark narrows to the core parameter classes.

### WP2: A parsimonious model of health-system escalation (M1–M28)

#### T2.1: Specify and identify the state model (M1–M9)

A Bayesian hierarchical Markov regime-switching model [Hamilton 1989] with an ordinal latent state (routine, elevated, strained, critical) observed through emergency calls, ED presentations and ICU occupancy; pre-specified weather, calendar and epidemic covariates; series-specific observation models sharing the state. Before real-data fitting: simulation-based identifiability and recovery; ordering constraints resolve label switching; if separation is insufficient, the fallback is an ordinal state-space formulation; the criterion is recovery of states and transitions, not visual fit. **Provisional state anchors**, fixed at registration and refined (not replaced) by T4.1's elicitation: **routine** below the 75th percentile of the seasonal baseline; **elevated** 75th to 90th; **strained** above the 90th, or sustained capacity pressure (ICU occupancy above 85%, ED boarding above its seasonal 90th percentile); **critical** above the 97.5th percentile, or capacity saturation. Anchors are percentile-based per event type; in estimation they enter as priors on state-dependent levels, not hard cutoffs.

#### T2.2: Represent the critical tail (M6–M14)

Peaks-over-threshold/generalised Pareto modelling for rare exceedances, coupled to the critical-state probability, threshold sensitivity reported: supporting, not a separate objective.

#### T2.3: Introduce evidence-derived priors (M10–M20)

Map WP1 distributions to the parameters where published evidence is relevant (weather effects, surge magnitudes, transition characteristics); compare weakly informative, fixed evidence-derived and adaptive robust borrowing. The adaptive specification is a robust mixture of an evidence-derived and a weakly informative component, prior-data conflict recorded explicitly. Power-prior and commensurate-prior approaches are sensitivity comparators.

#### T2.4: Add resilience indicators as a secondary information channel (M12–M20)

Rolling variance and lag-1 autocorrelation, entered as optional covariates on transition dynamics: do they add information beyond level/trend and the evidence prior? A null result is interpretable.

#### T2.5: Calibration, automation and validity monitoring (M20–M28)

Calibration methods suited to temporal dependence; a conformal component [Angelopoulos 2023; Barber 2023] as a robustness layer if simulation confirms its assumptions. This task also builds the **automation layer**: public streams (weather, surveillance, wastewater) ingested automatically as they become available; every model re-estimated at its inputs' native cadence (daily demand, weekly surveillance), with versioned automated re-fitting; and **assumption and validity diagnostics** on every fit (residual structure, dispersion, calibration) aligned with prediction-model standards [Collins 2015]: a model failing its checks is excluded by pre-registered rules, never silently retained. Release a documented reference implementation integrated with LiteRev-Evidence.

**Simulation is prior information, not data.** Fitting literature-parameterised trajectories as if data would count the same information twice and disable the conflict diagnostic H3b depends on;

simulation serves identifiability, structural constraints and prior predictive checks, and never tightens the evidence prior.

**Deliverables.** D2.1 model specification and identifiability study; D2.2 open implementation; D2.3 methodological paper.

**Risk.** Weak regime separation. *Mitigation:* simulation first; the ordinal fallback; report the identifiability boundary as a result rather than tuning until success.

*WP3: The decisive cold-start experiment (M12-M42)*

**T3.0: Outcome hierarchy, data-access gate and episode eligibility (M1-M14)**

Fixed before any evaluation is designed, never revisited in response to performance:

**Outcome.** The primary outcome is **daily respiratory-related emergency demand derived from CASU-144 records**, not a raw call count: "respiratory-related" is a pre-registered classification of recorded call reasons and urgency levels. **The heat arm scores heat-sensitive demand:** the same series and construction, restricted to cause classes fixed at registration (dehydration, renal, psychiatric), with respiratory-restricted and 75+ sensitivities. ED presentations and ICU occupancy enter as **additional observation channels on the shared latent state**; wastewater, sentinel consultations and weather as covariates; the hierarchy governs only which series H3a is scored on.

**Candidate outcome set and geography.** The pre-registered candidates, in hierarchy order: (1) cause-filtered CASU-144 call volume (primary); (2) ED presentations by category; (3) ICU occupancy; (4) all-cause 144 engagements and hospital admissions as sensitivity series; open surveillance (Sentinella, wastewater) as fallback outcomes only. The unit is the **canton of Geneva at daily resolution** (the HUG centrale's coverage area); extension to Vaud and Neuchâtel via the CHUV-operated 144 is a pre-specified external validation if access is granted.

**Data-access gate.** Criteria for **historical depth, resolution, latency and completeness** are fixed in advance; the primary outcome is selected at a pre-specified checkpoint (month 12) on those criteria alone.

**Episode eligibility.** An episode enters the confirmatory evaluation only if all of these are prospectively reconstructable: (1) a detectable onset under the prospective onset rule; (2) enough pre-onset history for its rolling baseline; (3) enough post-onset observations at the primary horizon; (4) the external evidence **as it stood at the historical origin**; (5) no leakage of future information; (6) separation from adjacent episodes. Failing episodes remain available for sensitivity analysis only. **At demand level, co-circulating pathogens form one episode** (concurrent influenza and RSV is one surge).

**Two registration points.** The hierarchy, gate criteria and eligibility rule are registered now. The window  $N$ , the archetype-specific horizons and the margin  $\Delta$  are registered after the checkpoint and the episode inventory, before any evaluation runs, by T3.3's pre-declared selection rules.

**Open-data validation track (M12-M24, pre-specified secondary analyses).** Before any clinical series connects, the complete rolling-origin machinery runs on open series: Swiss federal respiratory surveillance and wastewater, weekly all-cause deaths for the heat arm, and France's openly published daily emergency-activity indicators (OSCOUR ED attendances, SOS Médecins consultations), matching GESICA's Franco-Swiss frame. Archived multi-model hub forecasts provide contemporaneous external comparators for early, genuinely cold-start rounds [Cramer 2022; Sherratt 2023]. The track ships as a **public, re-runnable benchmark**, validates every component before

operational data arrive, and makes the central test robust to access outcomes; operational series then sharpen the claim to emergency-system demand.

### T3.1: Assemble the retrospective information set (*M12-M20*)

Harmonise the CASU-144 series with the additional channels and covariates; quantify completeness and delay; model right truncation and nowcasting [Höhle 2014; McGough 2020]; degraded reporting under strain could itself mimic an early-warning signal.

### T3.2: Reconstruct the true information set (*M18-M30*)

For each historical onset, create successive forecast origins using **only information available at that date. Admissibility as a prior is defined by referent, not venue**: prior inputs are statements about *other* populations, places or past events. **The confirmatory prior uses peer-reviewed literature and preprints only**, under the rolling cut-off. Timestamped situational reporting on the ongoing event *elsewhere* (WHO and ECDC situational reports) enters as a **pre-specified secondary prior variant**, through the same extraction and error pipeline, outside the confirmatory contrast: H3a tests literature borrowing, not dynamic evidence fusion. **Text describing the local event is excluded from any prior**: it is a noisy measurement of the outcome itself. Every input carries an index timestamp; origins re-run automatically through the T2.5 layer at each input's native cadence. Forecast horizons 7, 14 and 28 days (primary horizon per archetype fixed at the second registration point); the cold-start window counts local outcome observations **after a pre-defined real-time onset rule** that uses only variables available at the origin.

### T3.3: The model library and the automated selection engine (*M20-M34*)

**The library, registered before any evaluation, in two arms. Local-only**: seasonal GLM with weekday and holiday terms; persistence; established surveillance exceedance and nowcasting (Farrington/Noufaily, Bayesian nowcasting; operational implementations [Salmon 2016]); penalised, gradient-boosted and upper-tail quantile learners on the pre-specified covariates; the WP2 regime model with weakly informative priors. **Evidence-informed**: the same candidates equipped with WP1 priors and design inputs, in three registered variants: fixed evidence-derived priors; adaptive borrowing with conflict monitoring; adaptive borrowing plus resilience indicators (H3c). **Mechanistic transmission components [Keeling 2008] are admissible only for epidemiological archetypes**; environmental archetypes use exposure-response structures; admissibility per archetype is registered.

**Automated champion selection.** At each origin every admissible candidate runs automatically, passes the T2.5 assumption and validity checks, and is scored on pre-specified metrics (CRPS, log score, calibration); each arm's **champion** is selected by a registered rule, blind to the other arm. Identical machinery in both arms means the contrast isolates the evidence, not the selector; which family wins, where and when, is itself a result.

### Confirmatory testing procedure (fixed-sequence).

1. **Test 1 (primary).** Evidence-informed champion vs local-only champion, CRPS skill score, **respiratory** episodes, cold-start window  $N$ , at  $\alpha = 0.05$  two-sided. Passing requires paired-permutation  $p < \alpha$  *and* a lower confidence bound above the pre-registered minimal relevant improvement.
2. **Test 2 (generalisation).** The identical contrast on **heat** episodes, scored on heat-sensitive demand (T3.0), at the same  $\alpha$ , **conducted only if Test 1 passes**.

3. **If Test 1 fails**, H3a is not supported, Test 2 is exploratory, and the project's result is the failure map and the boundary condition.

All other contrasts (between library members, between borrowing strategies) are secondary. **The local-only arm is pinned in the registration**, its regime-model priors carrying a pre-declared band of vaguer and tighter alternatives reported as a sensitivity analysis: the comparator cannot be tuned into a straw man after the fact.

**Primary endpoint:** the CRPS skill score of the evidence-informed champion versus the local-only champion over the cold-start window. The unit of inference is the **episode** (origins are repeated measures within it); eligible episodes are few (13–14 respiratory candidates before screening), so inference is by **paired permutation over episodes**, block bootstrap alongside, both pre-specified, disagreement reported. **The design is powered by simulation-based operating characteristics at episode level** (type I error, power, interval width under realistic dependence and heterogeneity), run before the second registration point. The same simulation fixes, in advance, the **minimum number of eligible episodes** below which no confirmatory claim is made and the primary analysis becomes estimation, and the **selection rules** for  $N$  and  $\Delta$ :  $N$  the smallest grid value meeting the operating-characteristic criterion,  $\Delta$  set by a pre-declared rule on the simulated effect distribution, both chosen by algorithms blind to forecast performance. Secondary endpoints: log score, calibration (PIT, coverage), escalation detection at matched false-alarm rates. **H3b's  $\Delta$  is fixed here, before any evaluation:** non-inferiority is a CRPS deficit no greater than  $\Delta$ ; superiority is tested on T3.4's misspecified priors.

#### T3.4: Map benefit and failure (M28–M38)

Identify episodes where borrowing improves or worsens forecasts; characterise failure by population, outcome-definition, system, policy and temporal mismatch, extraction uncertainty and prior-data conflict, with deliberately misspecified priors as a stress test. The safety question: is harmful borrowing detectable early enough for adaptive discounting to limit its impact? The output is a **failure map**, not an average.

#### T3.5: Test generalisation (M34–M42)

The sequential heat test and, resources permitting, the legionellosis extension (year 4, not required for the central conclusion).

**Deliverables.** D3.1 reproducible cold-start evaluation pipeline; D3.2 primary result; D3.3 failure/stress-test map; D3.4 cross-archetype analysis.

**Risk: operational data access.** *Mitigation:* agreements initiated pre-award, letters accompany the application; and the open-data track already carries the methodological test under the same rolling-origin restriction, so delayed access narrows the claim (§2.4), it does not stall the project.

#### WP4: From predictive skill to operational value (M24–M48)

##### T4.1: Elicit operational losses and thresholds (M24–M32)

Structured **SHELF** elicitation ( $n \approx 15$ –20 across HUG emergency medicine, CASU-144 regulation and capacity management): elicit the consequences of early, late and unnecessary escalation, then derive thresholds from the losses. **The primary decision is fixed in advance:** trigger surge-capacity escalation when the forecast probability of the strained/critical state crosses the elicited threshold.

#### T4.2: Decision-analytic evaluation and equity audit (M30-M40)

Re-evaluate the WP3 forecasts with net-benefit/decision-curve analysis and value-of-information: do rankings change once consequences are incorporated? Because operational records may encode structural differences, assess calibration and threshold performance across aggregate strata (age, sex, neighbourhood deprivation where appropriate): calibrated on average but miscalibrated for a relevant group is not operationally ready.

#### T4.3: Counterfactuals, observation-mode dashboard and living updating (M34-M48)

Estimate, for selected episodes, what would have changed had escalation followed the model's signal (a simple capacity model, propagated uncertainty: counterfactuals, not causal estimates). For event types whose models pass T3.3 validation and T4.2's decision-value check, and subject to authorisation, the framework then runs as a **daily observation-mode dashboard** with the partner services (CASU-144 regulation, HUG emergency department, ICU capacity management): state probabilities, forecasts and validity checks refreshed automatically, **recorded, never used clinically**. A **living-evidence monitor** re-runs the per-scenario reviews [Elliott 2014] and flags new publications that would change a parameter value, a variable set, an outcome threshold or the recommended model; proposed changes apply only through version-controlled, pre-registered update rules, never silently. If authorisation is not granted, the project is complete on retrospective evaluation and says so.

**Deliverables.** D4.1 elicited loss structure and equity audit; D4.2 decision-analytic evaluation; D4.3 counterfactual analysis, with the observation-mode dashboard and living-update protocol where authorised.

**Methods, data protection and reproducibility.** Version-controlled R/Python and a registered analysis plan; clinical data processed in the UNIGE secure environment under a new CCER approval with me as applicant; benchmark and software released openly, synthetic equivalents where possible.

**Expected outputs.** Four to six papers and the durable open resources of \$2.5; **I lead the methodological, benchmark and integrative outputs.** None of the explicit fallbacks above converts an inconclusive analysis into a success claim.

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#### 2.3.3 Research environment, team and resources

**Host: Institute of Global Health, Faculty of Medicine, University of Geneva**, with a methodological collaboration with **Data Science for Digital Health** (Prof. Douglas Teodoro) and a clinical collaboration with **HUG emergency medicine** (Prof. Thibaut Desmettre; Dr Robert Larribau, médecin responsable CASU-144).

**Institute of Global Health:** epidemiology, infectious-disease modelling and automated evidence extraction in one institute: where the gap became visible. Independence is organisational: a programme led by me alongside the Institute's groups, not inside one (\$2.6). **DS4DH** contributes the biomedical NLP WP1 needs; through GESICA I already work with the emergency-medicine partners; the AI-in-EMS review [Edjinedja 2026] evidences the collaboration.

**Existing infrastructure.** LiteRev-Evidence is a prototype (live federated search of the open literature, screening, living reviews, certainty-graded extraction with provenance, pooling into parameter distributions, candidate variables, typed outcome templates, multi-family model fitting with automated tuning, weather and surveillance connectors); WP1 and WP2 turn it into a validated instrument and test it. The legionellosis study (BASEC 2026-00324, ethics granted) supplies the

contrasting archetype. **Computing:** UNIGE HPC (Baobab/Yggdrasil) and the University's secure environment for clinical data.

#### Commitments, by status.

| Item                                                    | Status         | Evidence / action                                                                                                                                          |
|---------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Legionellosis linked data                               | <b>Secured</b> | Ethics approval BASEC 2026-00324                                                                                                                           |
| Host institute (ISG)                                    | Requested      | Host confirmation, SNSF template                                                                                                                           |
| DS4DH methodological collaboration                      | <b>Agreed</b>  | Letter requested                                                                                                                                           |
| Institutional confirmation                              | Requested      | General confirmation, UNIGE Vice-Rectorate                                                                                                                 |
| <b>CASU-144 records (HUG-operated), primary outcome</b> | Requested      | Letter requested; continuous, daily, near real-time; archive depth to confirm; <b>HUG access established within GESICA</b> ; own agreement + CCER required |
| ED presentations + ICU occupancy, additional channels   | Requested      | Letters requested; daily depth to confirm; ICU fallback in WP3                                                                                             |
| Operational-data ethics                                 | Requested      | CCER submission, PI as applicant                                                                                                                           |
| Computing                                               | Requested      | UNIGE HPC access                                                                                                                                           |

The three operational-data rows are the key remaining feasibility items.

**Resources requested.** Project funds are capped at **CHF 250,000 over four years**; the applicant's salary is covered separately. Doctoral students and postdocs cannot be employed (2026 rules); support personnel can.

| Item                                                         | Rationale                                                                            |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Scientific/technical collaborator, <b>50% over 48 months</b> | WP1 benchmark extraction, WP3 harmonisation and evaluation pipeline, reproducibility |
| Second independent extractor                                 | Contracted M3-M9; required for T1.2's dual-extraction design                         |
| Computing and data access                                    | Evidence processing, Bayesian estimation, rolling-origin evaluation                  |
| Travel, conferences and incoming visits                      | Results presentation and sustained international collaboration                       |
| Other eligible direct costs                                  | As justified in the final SNSF budget                                                |

**Preparatory work before the grant starts.** Advanced before month 1, outside grant funding: **data agreements** with HUG and 144/CASU; **CCER preparation with myself as applicant**; **infrastructure hardening** of LiteRev-Evidence before clinical data connect. Feasibility rests on one structural property (**the prototype already exists**) and on the fallbacks, each leaving the central question answerable with the narrowing stated.

## 2.4 Schedule and milestones

**Who does what.** I execute WP2 and WP3's confirmatory design personally and lead WP1's protocol and WP4's decision work. One **scientific/technical collaborator** (support personnel, 50% over 48 months) carries three fully specified tasks: T1.2's second independent extractor, the T3.1-T3.2 harmonisation and automated rolling-origin pipeline, and the reproducibility engineering behind D1.1 and D3.1. No other personnel are requested.

The design avoids a serial chain: WP2 falls back to weakly informative priors if WP1 finds extraction inadequate, WP3's open-data track runs regardless of operational access, and WP4's decision analysis is retrospective. **What falling back to open data alone costs:** open series are not emergency-system demand, so the outcome claim narrows and most of WP4 turns illustrative; hence the agreements are a pre-award action.

| Milestone | Month | Criterion                                                                         |
|-----------|-------|-----------------------------------------------------------------------------------|
| M1        | 9     | Extraction benchmark released                                                     |
| M2        | 12    | Regime/state representation passes identifiability criteria, or fallback selected |
| M3        | 20    | Open-data benchmark running; operational series harmonised, or fallback activated |
| M4        | 34    | Primary cold-start hypothesis tested by the registered champion contrast          |
| M5        | 40    | Decision relevance established                                                    |
| M6        | 48    | Cross-domain validation; observation-mode dashboard running where authorised      |

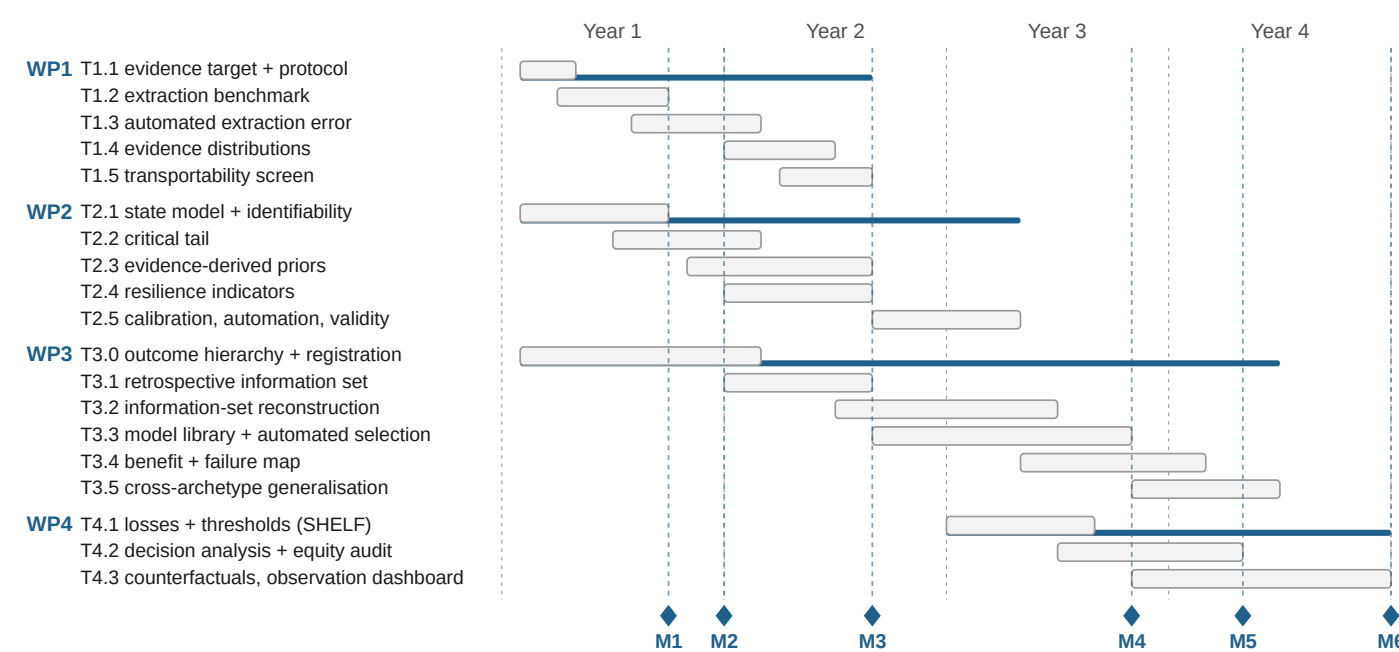


Figure 2: work packages, tasks and milestones over 48 months.

### 2.5 Relevance and impact

The result is useful either way: a principled, auditable way to borrow if evidence helps; a stated boundary if not.

**A null result changes identifiable practice.** Modellers routinely seed early-crisis models with parameters lifted from a few studies, and evidence-synthesis platforms (including my own) assume pooled estimates are reusable as priors; a calibrated null tells those groups the practice buys nothing where it is most used.

**Form of publication.** Peer-reviewed articles, preprints on submission, open access throughout; benchmarks, prior library and code as **citable, versioned open resources**; results returned to the partners.

Durable resources remain, subordinate to the central question: **the extraction and replication benchmark, the public cold-start benchmark on open series**, an **open reference framework** for evidence-informed latent-state forecasting, and a **decision-analytic evaluation framework**.

**Practical and societal impact.** In a crisis's first weeks, emergency systems decide whether to open capacity, redistribute resources or escalate before local outcomes provide a reliable base; the project quantifies whether external evidence improves those decisions, and with how much residual uncertainty. Geneva anchors it; the contrasting domains show what transfers.

**Where this points operationally.** Each validated event type leaves a registered, automatically re-validated forecasting configuration and a daily observation-mode dashboard in the hands of the services that would use it. Extending that nucleus into a multi-crisis surveillance platform for clinicians and regulators, event type by event type as models validate, is the explicit post-grant pathway; this project supplies the validated core and update rules that make it trustworthy.

## 2.6 Relevance for personal career development

**Why I am ready, and why now.** I have the methods, the prototype and the domain access, used so far inside other people's programmes; what I have not had is a research agenda that is mine to set. None of the programmes below has this project's objective.

| Programme                               | Objective                                                                        | My role              | Distinction from Ambizione                                       |
|-----------------------------------------|----------------------------------------------------------------------------------|----------------------|------------------------------------------------------------------|
| GESICA                                  | Franco-Swiss crisis intelligence                                                 | Contributor          | Decision support; no test of evidence-derived priors             |
| GeoAI4EI<br>(Horizon Europe, 2026–2030) | European epidemic-intelligence toolbox                                           | Contributor          | Infrastructure; no test of borrowed evidence's inferential value |
| Legionellosis study (ongoing)           | Geneva waterborne outbreak determinants                                          | Data/scientific lead | Supplies a contrasting validation extension                      |
| <b>Ambizione</b>                        | <b>When evidence-informed forecasting helps at cold start, and when it harms</b> | <b>PI, full-time</b> | -                                                                |

**This project does not build another epidemic-intelligence platform.** It uses existing infrastructures to test a question that is a work package of none of them.

**The question and the methods are mine.** It arose building LiteRev-Evidence: the platform could pool published estimates into priors, but nothing established whether those priors improved forecasting in a new setting. The methodological core came from fifteen years of quantitative finance, from no group I have worked in; the platform is UNIGE intellectual property under my scientific direction.

**Collaborators and their roles.** Prof. Olivia Keiser leads the group in which I developed LiteRev, the MAS on which I teach, the legionellosis study and GeoAI4EI; she co-authors my doctoral and post-doctoral output (the expected profile at this stage) and has **no scientific role in this project**. Prof. Douglas Teodoro leads GESICA and provides the WP1 biomedical-NLP collaboration.

**The organisational position is the substance of the transition.** Hosted by the **Institute of Global Health**, I will lead an independent programme alongside the Institute's groups rather than inside one; the PI's independence guarantees are documented in the confirmation letters. The

programme is executable by the PI and one budgeted collaborator: **no named individual is load-bearing, by design.**

**Where this leads.** By month 48 I intend to hold a validated answer, open resources the field can build on, and a first-author methodological record in my own name: the basis for a group-leadership position or professorship. My research time is currently divided across four collaborative programmes; Ambizione consolidates it into one.

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